

Unit

1

Lesson

2

Spark The Challenge!

Innovate to Elevate.

Time Frame: [90 min].

Lesson Objectives:

By the end of this lesson students will be able to:

- Identify problems related to waste sorting in school environments.
- Understand the role of a Recycling Technician.
- Use the Engineering Design Process (EDP) to solve a real-world problem.
- Brainstorm and plan a smart solution.
- Recognize how recycling helps the environment and community.

Challenge questions of the lesson

1. What could make it easier for everyone in the school to sort their trash correctly?
2. If you were a Recycle Technician, what new idea or system would you add to help our school manage trash better?
3. What part of your waste management idea or design helps make our school cleaner and helps everyone sort their trash properly?

SEL Relation

- **Community Awareness (Understanding our world):** We will learn about waste management systems in our community and appreciate the vital role of professionals like the Waste Management Specialist.
- **Responsibility (Doing our part):** We will understand that everyone, including ourselves, has a responsibility to properly sort trash, recycle, and keep our environment clean.
- **Collaboration (Working together):** By sharing our ideas for waste reduction and recycling, we can work together to create a cleaner and more sustainable environment.
- **Appreciation (Feeling thankful):** We can learn to appreciate the efforts of those who manage our waste and the resources we save by recycling and repurposing.
- **Self-awareness:** Recognizing what they learned and what they can improve in their understanding of waste management and their role in it.
- **Communication:** Expressing ideas clearly through speaking, drawing, or writing about waste solutions and recycled projects.
- **Critical Thinking & Problem-Solving:** Analyzing the problem of waste, evaluating different sorting methods, and developing creative solutions for repurposing materials.
- **Environmental Literacy/Responsibility:** Understanding environmental issues related to waste and taking action to protect the planet through recycling and reuse.

Engage



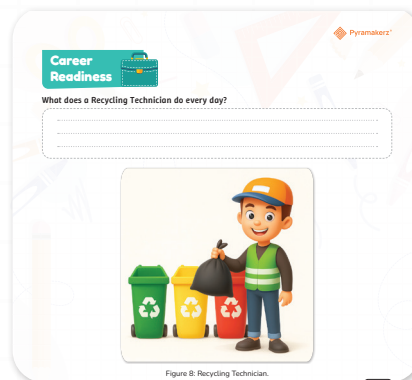
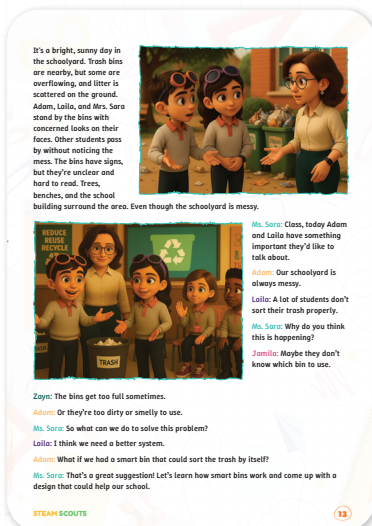
Teacher Role:

- **Storyteller:** Begin by reading or narrating a story that presents a real-world problem (e.g., a messy schoolyard). Use tone, visuals, and body language to engage students emotionally.
- **Bridge Builder:** Connect the problem in the story to the real world by saying, "Just like Adam and Laila want to solve the messy schoolyard problem, there are people in the world whose job it is to help communities manage trash and keep our planet clean!"
- **Discussion Leader:** Ask reflection questions to prompt students to think about the problem.

Expected From Students:

- **Active Listening:** Students will listen carefully to the story and pay attention to the problem faced in the schoolyard.
- **Observation:** Students will notice details about the messy schoolyard.
- **Participation:** Students will contribute to classroom discussion by sharing their initial ideas or reflections.

Parts of the books included.



Story



1. Start with a Warm Welcome and Story Recap:

- “Good morning, engineers! Last session, we started a story about our friends Adam and Laila and the messy schoolyard. Do you remember what happened? Who remembers what problem they noticed?”

- (Allow a few responses to activate memory and emotional connection, focusing on the messy schoolyard problem from the provided image/dialogue.)

2. Continue the Story with Expression:

- “Well, today is a brand-new day in their classroom. The sun is shining... and something exciting is about to happen. Let’s see what Adam and Laila decided to do!”
- (Read the story continuation aloud with tone and character voices, referencing the dialogue and scene)

Possible Strategies for Implementation:

Turn and Talk: “What Would You Do?”

- **Prompt:** “If you were in Mrs. Sara’s class, what kind of 'better system' or 'smart bin' would you design for our schoolyard to help with the messy trash problem?”
- (Let students discuss ideas with a partner for 2 minutes).

Career Readiness



Objective:

Students will understand the role of a Recycling Technician by identifying their main responsibility ensuring that waste is properly managed and that spaces are clean and functional for everyone. This connects the lesson to a real-world career that focuses on environmental responsibility and innovative problem-solving.

Possible Answer:

What does a Recycling Technician do every day?

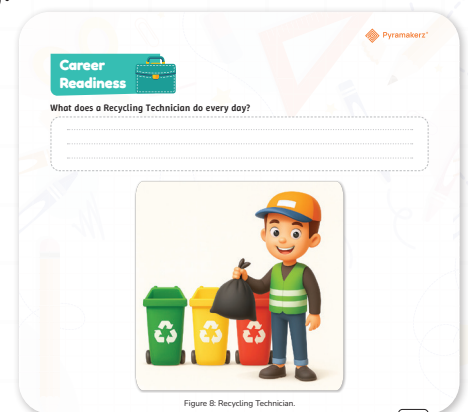
A Recycling Technician helps sort waste into the right bins, checks machines that recycle materials, and makes sure that plastic, paper, and metal are reused to protect the environment.

Strategies for implementation:

1. Think-Pair-Share with Reflective Reveal:

Strategy: "What's Recycling Technician?"

- How to Implement:
 - Begin by asking: "What do you think a Recycling Technician does every day?"
 - Students first think quietly for 1 minute.
 - Then, they pair up to discuss their ideas.
 - Invite 2–3 pairs to share out with the whole class.
 - Enhancement:
 1. Before revealing the correct answer, tell students: "Let’s watch a short video to see if our ideas were correct or if we’ll learn something new about these important jobs!"



2. **Play the video:** <https://www.youtube.com/watch?v=XR6wkDG4Gzg>
3. After watching, ask: "Did your answer change? What did you learn from the video about how they help manage waste and keep places clean?"



(This strategy supports reflection, listening, and correcting misconceptions through student-centered learning.)

2. Role-Play with Real-World Connection:

Strategy: "The Schoolyard Challenge: Expert Consultation"

- How to Implement:
 - Divide students into small groups, designating one student in each group as the "Recycling Technician" and the others as the "Schoolyard Design Team" (like Adam and Laila).
 - Present the Challenge: "Your schoolyard still has some trash problems, and your job as the Design Team is to make it better! The Recycling Technician in your group is here to help you come up with the best solutions."
 - Specialist's Role: Provide each "Specialist" with a small card or list of "Expert Tips" (these are the same tips as before, but framed as advice).
- **Expert Tips:**
 1. "Think about clear signs and labels for every bin."
 2. "Consider how often the bins will be emptied so they don't overflow."
 3. "Make sure the bins are easy for everyone to use, even smaller students."
 4. "Explore what materials can be reused, recycled, or even composted for new things."
 5. "How can we make sorting fun and easy for everyone?"
- Design Team's Role: Encourage the "Design Team" to ask their "Specialist" questions about their ideas for improving the schoolyard's waste system ("What kind of bin should we use for paper?", "How can we stop trash from blowing away?").
- Group Brainstorm & Dialogue: Allow 5-7 minutes for groups to discuss their ideas, with the Specialist guiding them using the tips.
- Share Out: Bring the class back together and have each "Design Team" share one problem they discussed and the solution their "Specialist" helped them develop.

Explain



Teacher Role:

- **Activity Introducer:** Clearly present the video, explaining that it will show how engineers solve problems, which can help them design their own "smart trash bin."
- **Purpose Setter:** Explain what students should look for in the video, emphasizing the problem-solving steps (EDP) and how they relate to making a "smart trash bin."
- **Engagement Encourager:** Prompt students to actively observe and think critically while watching, relating the video content to solving the messy schoolyard problem.
- **Discussion Moderator:** Lead the post-watching discussion, ensuring all students have an opportunity to share their observations and insights about the EDP.
- **Concept Clarifier:** Address any questions or misunderstandings that may arise about the EDP steps and how they apply to waste management solutions.
- **Connection Maker:** Help students connect the information in the video to their previous discussions about the messy schoolyard and their own ideas for a "better system" or "smart bin."

Expected From Students:

- **Active Viewing:** Students will watch the video attentively, focusing on identifying the steps of the Engineering Design Process (EDP) as presented.
- **Observation & Analysis:** Students will recognize how problem-solving steps are used in the video to create a solution.
- **Thoughtful Participation:** Students will contribute meaningfully to discussions, sharing their observations and reflections on the EDP.
- **Application of Concepts:** Students will begin to connect the EDP steps to the challenge of designing a "smart trash bin" for their school.

Parts of the books included.



Screen Lessons

Objective:

Students will understand how the Engineering Design Process (EDP) is used to solve problems, specifically in the context of creating solutions for Recycling Technician, by actively engaging with video content and applying the EDP steps to a real-world scenario.

Video link: <https://www.youtube.com/watch?v=fxJWin195kU>





Screen Lessons

"Now it's time to be a helper and use our problem-solving steps (EDP) to make the smart trash bin."

<https://www.youtube.com/watch?v=fxJWin195kU>



Possible Strategies for implementation:

1. Pre-Watching: Problem Solvers in Training!

Strategy: "What's an Engineer?"

- How to Implement: Begin by asking: "We've been talking about the messy schoolyard and thinking about smart bins. Who do you think helps solve big problems like this and creates new things? These people are called engineers!"
- "Today, we're going to watch a short video about 'The Engineering Process' to see how engineers think and solve problems. Pay close attention to the steps they follow!"
- Guide students to the video link: https://www.youtube.com/watch?v=MAhpfFt_mWM

2. During Watching: Spot the Steps!

Strategy: "EDP Detective Checklist"

- How to Implement: Provide students with a simple checklist of the EDP steps (Ask, Imagine, Plan, Create, Improve) (in the main book). As they watch the video, encourage them to "check off" or make a quick note when they see an example of each step being used.



3. Post-Watching: Our Engineering Brains!

Strategy: "Connect the Steps to Our Schoolyard"

- How to Implement: Lead a discussion reviewing the EDP steps from the video.
- Ask questions like:
 1. "What was the first step the engineers did in the video?" (Ask).
 2. "How did the engineers 'Imagine' different solutions?"
 3. "What did they have to 'Plan' before they started building?"
 4. "What did they 'Create'?"
 5. "Why was 'Improve' an important step?"
- "Now, let's think about our messy schoolyard. How could we use these engineering steps to design our smart trash bin?"
 1. "What would we 'Ask' about the problem?"
 2. "What could we 'Imagine' for a solution?"

Engineering Design Process

Name:

Project:

Ask

What is the problem we are trying to solve?
Our schoolyard is messy because students don't throw trash in the right bins. Sometimes the bins are full, dirty, or confusing. We want to build a smart bin that sorts trash correctly and tells us when it's full.

Who has the problem?
Mr. Yusef (the janitor), the students, and the school staff.

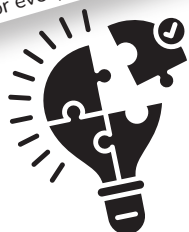
Why is it important to solve this problem?
It helps keep the school clean, protects the environment, and makes waste sorting easier for everyone.



Imagine

What are some ways to solve the problem?
Use sensors to sort wet and dry waste automatically.
Add a lid that opens without touching.
Include lights or a screen to show if the bin is full.
Power the bin with solar energy in case there's no electricity.

What materials or tools might we need?
Arduino, moisture sensor, IR sensor, servo motor, LED lights, LCD screen, wires, Tinkercad for design, and cardboard for a DIY model.



Plan

What will your smart bin look like?
Sketch or describe it.

Our smart bin will have two compartments: one for wet waste and one for dry waste. It will have a lid that opens automatically with a servo motor and IR sensor. LED lights or a small screen will show if it's full.



Create





Mastermind

Objective:

Students will creatively apply their understanding of waste reduction and reuse by transforming an old or unused item from home into something new and useful. This activity promotes innovative thinking, resourcefulness, and a personal connection to environmental stewardship.

Activity Instructions

Step 1: Introduce the Mission (in class). Say to students:

- "You've learned so much about recycling and sorting trash, but what if we could give something a brand-new life before it even gets to a bin? Your mission today is to become a 'Recycling Mastermind' by transforming something old into something new!"
 - Show the "Bring It Back to Life" graphic or explain the concept of reusing items instead of throwing them away.
 - Briefly review the steps involved: Find, Think, Create, Share.

Step 2: The "Find" and "Think" Challenge (at home preparation).

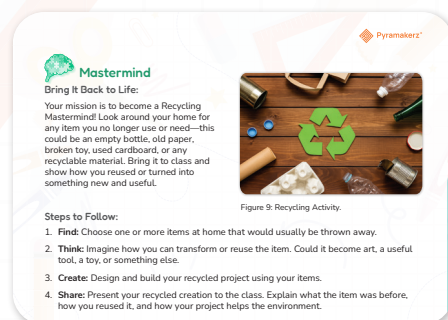
- "Find": Explain to students: "Your first step is to look around your home for any item you no longer use or need. This could be an empty bottle, old paper, a broken toy part, used cardboard, or any recyclable material. Ask a grown-up for help to find something!"
- "Think": "Once you find your item, 'Think' about how you can transform or reuse it. Could it become a piece of art, a useful tool, a new toy, or something else entirely? Be creative!"
- Home Connection: Send a note home to parents explaining the "Bring It Back to Life" project, the types of items to look for, and encouraging family brainstorming.

Step 3: "Create" Your Masterpiece (in class).

- "Today in class, you will 'Create' your recycled project! You'll design and build your new creation using the item you brought from home."
- Provide a variety of common craft supplies (glue, tape, scissors, markers, extra paper, fabric scraps, etc.)
- Circulate around the room, offering guidance and encouraging students as they work on their projects.

Step 4: "Share" Your Transformation (in class presentation).

- "Now it's time to 'Share' your amazing recycled creation with the class!"
- Tell students they will present their project and explain:
 - What the item was before they reused it.
 - How they reused it or turned it into something new and useful.



- How their project helps the environment ("I saved this from the trash!", "This means we didn't have to make something new!").
- Facilitate presentations, allowing time for questions and applause.

Possible Strategies for implementation:

1. "Show & Tell Prep Stations" (Guided Presentation Practice):

- How to Implement:
 - Provide simple sentence starters or prompts on cards at small group stations ("My item was a _____. Now it is a _____. I helped the Earth by _____."). Students can practice saying these sentences with their project to a partner before presenting to the whole class.

2. "Reuse Idea Gallery" (Inspiration and Brainstorming):

- How to Implement:
 - Before students bring their items, show a digital or physical "gallery" of simple upcycled creations made from common household items (e.g., a toilet paper roll telescope, an egg carton planter). This can spark ideas and reduce anxiety about finding "the perfect" item.

3. "Environmental Impact Tags" (Connecting to the 'Why'):

- How to Implement:
 - Provide small tags or labels for students to attach to their finished projects. On the tag, they can draw a picture or write a word/phrase about how their project helps the environment (e.g., "Less Trash!", "New Life!", "Save Trees!"). This reinforces the objective of the activity.



Now I Can

Objective:

Students will self-assess and reflect on their learning journey, recognizing their ability to contribute to solving waste management challenges. Using a "Now I Can" checklist, they will demonstrate their understanding of generating ideas, the engineering design process, the importance of a Recycling Technician's role, and the value of planning in problem-solving. This activity supports the development of metacognition, confidence, and ownership of learning in a student-friendly, age-appropriate way.

Activity Instructions:

Step 1: Create a Calm, Supportive Reflection Moment

- Begin with: "Let's take a few minutes to think about everything you learned and how you became a real environmental engineer and problem-solver for our school's waste system!"
- Set a positive tone.

Step 2: Introduce the "Now I Can" Checklist

- Show the checklist and read each item aloud slowly, one at a time.

- After each one, pause and ask:
 - "Can you give me a thumbs up if you can do this now?"
 - "Can someone share how they did this during our project?" ("I generated ideas when we talked about the smart bin!").
 - This supports oral reflection and includes students who may not write confidently yet.

Step 3: Reflect with Drawings or Sentences

- Pass out the checklist with space for them to check off.
- Ask students to draw or write one thing they are proud of: "What is something you thought, said, or learned that helped us think about a cleaner school or planet?"
- This gives a visual and personal way to reflect.

Step 4: Partner Sharing

- Prompt students to turn to a partner and say:
 - "One thing I learned about solving problems for our environment is..."
 - "Next time, I would like to try..."
- Develops speaking and listening skills.

Step 5: Celebrate Growth

- Say: "Look at all you've done! Your real environmental problem-solvers who think about keeping our Earth clean!"
- You may give a simple "I'm an Environmental Engineer" badge or sticker.

Relation to Standards.

1. NGSS (Next Generation Science Standards)

- **K-LS1-1 (Life Science - From Molecules to Organisms: Structures and Processes)**
 - **Standard:** Use observations to describe patterns of what plants and animals (including humans) need to survive.
 - **Relation:** Understanding that proper waste management (reducing pollution, keeping environments clean) contributes to healthy ecosystems and human well-being for survival aligns broadly with environmental health.
 - **In this lesson:** Students understand that a clean environment (less trash, more recycling) is better for plants, animals, and people.
- **K-ESS3-3 (Earth and Space Sciences - Earth and Human Activity)**
 - **Standard:** Communicate solutions that will reduce the impact of humans on the land, water, and/or air.
 - **Relation:** This standard is directly aligned with the core focus of the unit. Students learn and communicate solutions for waste reduction and recycling.
 - **In this lesson:** Students design sorting bins, sort waste, explain the importance of recycling, and create posters communicating solutions to reduce human impact (trash) on the environment. They also "Bring It Back to Life" by reusing items,

reducing waste.

- **K-2-ETS1-1 (Engineering Design - Defining and Delimiting Engineering Problems).**
 - **Standard:** Ask questions, make observations, and gather information about a situation people want to change to define a simple problem.
 - **Relation:** The initial story of the messy schoolyard explicitly sets up a problem-defining scenario.
 - **In this lesson:** Students identify the "messy schoolyard" as a problem, ask "why does this happen?", and observe unsorted trash to define the need for a "better system" or "smart bin". Students reflect on their ability to generate ideas for improving the school's waste system.
- **K-2-ETS1-2 (Engineering Design - Developing Possible Solutions).**
 - **Standard:** Develop a simple sketch, drawing, or physical model to illustrate how the shape of an object helps it function.
 - **Relation:** The activity where students design "Sorting Stations" directly involves developing solutions through drawings or models. The "Bring It Back to Life" task involves creating a new item from an old one.
 - **In this lesson:** Students design visual solutions for sorting bins and illustrate how labels/colors function to improve waste separation. They also design and build recycled projects, demonstrating how a new form serves a new function. Students recall their design ideas and explain what they planned.
- **K-2-ETS1-3 (Engineering Design - Optimizing the Design Solution).**
 - **Standard:** Analyze data from tests of two objects designed to solve the same problem to compare the strengths and weaknesses of how each performs.
 - **Relation:** The comparison between current messy systems and proposed "smart bins" or improved sorting methods reflects analyzing potential solutions. The "DIY Lab" (Wordwall game) involves 'testing' their sorting knowledge.
 - **In this lesson:** Students implicitly compare the "messy schoolyard" (current system) with ideas for a "smart bin" or better sorting (improved system) and reflect on how effective different

2. Common Core – ELA Standards

- **CCSS.ELA-LITERACY.SL.1.1 (Speaking & Listening).**
 - **Standard:** Participate in collaborative conversations with diverse partners about grade 1 topics and texts.
 - **Relation:** This standard is met through various interactive activities.
 - **In this lesson:** Students engage in "Think-Pair-Share" discussions, "Consulting Our Expert" role-play, partner sharing during reflection, and group discussions about waste sorting.
- **CCSS.ELA-LITERACY.W.1.8 (Writing).**
 - **Standard:** With guidance and support, recall information from experiences or gather information to answer a question.
 - **Relation:** Students recall and reflect on their project experience in both drawing and

writing throughout the unit.

- **In this lesson:** Students write observations about the Waste Management Specialist, draw or write about their "Now I Can" learning, and explain their recycled creations.

3. ISTE – Technology Standards for Students.

- **Empowered Learner (1a, 1c).**

- **Standard:** Students articulate and set personal learning goals, develop strategies to achieve them and reflect on the learning process. Students use technology to seek feedback that informs and improves their practice and to demonstrate their learning in a variety of ways.
- **Relation:** Students actively engage with digital tools (Wordwall game, videos), reflect on their learning via "Now I Can", and demonstrate understanding through various outputs (designs, posters, answers, repurposed items).
- **In this lesson:** The "Now I Can" checklist allows students to show what they learned and how they contributed, encouraging self-awareness and ownership of their learning goals and the process.

- **Innovative Designer (4a, 4b, 4c).**

- **Standard:** Students know and use a deliberate design process for generating ideas, testing theories, creating innovative artifacts or solving authentic problems. Students select and use digital tools to plan and manage a design process that considers design constraints and anticipated effects. Students develop, test and refine prototypes as part of a cyclical design process.
- **Relation:** The unit involves identifying a problem (messy schoolyard), brainstorming solutions (smart bins, sorting), designing solutions (bin labels, posters), and creating artifacts (repurposed items), directly reflecting the EDP.
- **In this lesson:** Students identify the problem of unsorted trash, generate ideas for improving the waste system, design a "Sorting Station", create educational posters, and transform items through a design process. They also learn the steps an engineer uses to solve a design challenge.

- **SEL & 21st Century Skills.**

- **Self-awareness:**

- **Relation:** Recognizing what they learned and what they can improve.
- **In this lesson:** Students use the "Now I Can" checklist to reflect on their accomplishments and areas for growth.

- **Communication:**

- **Relation:** Expressing ideas clearly through speaking, drawing, or writing.
- **In this lesson:** Students present their repurposed projects, discuss ideas in groups, contribute to class discussions, and design informative posters.

- **Collaboration:**

- **Relation:** Working effectively with others to achieve a common goal.

- **In this lesson:** Students participate in "Think-Pair-Share" activities, "Consulting Our Expert" role-play, and group discussions during sorting activities.
- **Critical Thinking & Problem-Solving:**
 - **Relation:** Analyzing information, evaluating ideas, and developing solutions to challenges.
 - **In this lesson:** Students identify the messy schoolyard problem, brainstorm solutions for waste sorting, apply the Engineering Design Process, and creatively transform items.
- **Environmental Literacy/Responsibility:**
 - **Relation:** Understanding environmental issues and taking action to protect the planet.
 - **In this lesson:** The entire unit focuses on understanding waste management, the importance of recycling and reusing, and the role of community members and professionals in environmental protection.